Braeden Burger

(810) 835-5064 | beburger@umich.edu | https://www.linkedin.com/in/braeden-burger/
https://github.com/2002goblue/

Objective

Motivated Computer Science student in the BS/MS program at the University of Michigan–Flint with strong skills in C++, Python, Java, and C#. Seeking a **Software Development Internship** to contribute to the design, implementation, and testing of reliable software solutions. Experienced in applying principles of **software engineering**, **AI integration**, and **human-computer interaction** to create efficient, user-focused applications. Proven ability to collaborate in multidisciplinary teams, debug complex systems, and deliver high-quality results through iterative development and data-driven problem solving.

Education

University of Michigan-Flint | Flint, MI
Bachelor of Science in Computer Science (4.0 GPA) Expected Apr. 2026
Master of Science in Computer Science and Information Systems (SWE Concentration) Expected Apr. 2027

Work Experience

University of Michigan-Flint | Flint, Michigan Aug. 2024 - Present
Research Assistant

- Implementing and debugging applications in **C#** and **Python**, integrating **AI** and **networking components.**
- Work collaboratively with a multidisciplinary team to design, implement, and evaluate complex software systems.
- Conducting qualitative and speculative HCI research on digital consent, cross-reality interaction, and social VR.
- **First author of a full paper** that was accepted and presented at the 2025 ACM CHI conference in Tokyo, Japan.
- **Co-author and presenter of a full paper** accepted to the 2025 ACM CSCW conference in Bergen, Norway.

National Science Foundation (REU) | Flint, Michigan May 2025 - Aug. 2025
NSF REU Researcher (Internship)

- Designed and built an **XR interaction application** in **C#** for smart glasses using real-time data processing.
- Integrating **generative and multimodal AI** in XR glasses, enabling context-aware interactions in social settings.
- Developed **in-depth user studies** aimed at informing future approaches to safer and more intuitive XR design.
- Managed project timelines and documentation while iterating on designs through usability testing and feedback.

National Science Foundation (REU) | Flint, Michigan May 2024 - Aug. 2024
NSF REU Researcher (Internship)

- **Managed a team of researchers** in the development of an innovative **Bluetooth-connected haptic device.**
- Used **C++** and **embedded firmware** to enhance communication of individuals with hearing and visual disabilities.
- Analyzed user testing data to evaluate haptic device performance and guide **iterative design improvements.**

University of Michigan-Flint | Flint, Michigan Oct. 2023 - Sep. 2024
Undergraduate Researcher (UROP)

- Analyzed and categorized speculative qualitative data to investigate emerging interaction concepts in social VR.
- **Co-authored a CHI workshop paper** and participated in the **ACM CHI 2024** workshop in Honolulu, Hawaii.
- **Co-authored a short paper accepted to CSCW 2024** exploring consent and safety in cross-reality spaces.
- Presented findings at UM MISC Research Showcase and the College of Innovation & Technology Project Day.

Technical Skills

- **Languages:** C++, Python, Java, C#, JavaScript, HTML/CSS
- **Frameworks/Tools:** VS Code, GitHub, Linux, Node.js, MongoDB, Arduino
- **Concepts:** Data Structures and Algorithms, Computer Networking, XR UI/UX
- **Research & Design:** Human-Computer Interaction, User-Centered Design, Data Analysis, Qualitative Research

Projects

University of Michigan-Flint | Flint, MI
DrillVision - AI-Powered Marching Band Drill Analyzer Aug. 2024 - May 2025

- Developed with **Python, PyTorch, OpenCV, Unity, Node.js, AWS**
- **Developed a computer-vision system** to detect and analyze marching band formations using Faster R-CNN.
- Implemented IoU/mAP metrics for validation and **trained models with custom datasets of band images.**
- Deployed backend on **AWS EC2** using **Node.js** for model inferencing, data processing, and API handling.
- Configured **AWS S3** for secure storage and retrieval of images, datasets, and processed model outputs.

Andrew Haralambous
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September 9, 2025

Hiring Committee
College of Innovation and Technology
University of Michigan-Flint
303 E Kearsley St
Flint, Michigan 48507

Dear Hiring Committee,

My name is Andrew Haralambous, and I'm a junior BS/MS student at the University of Michigan-Flint, studying a BS in Computer Science and an MS in Software Engineering (Cybersecurity Engineering). I am writing to express my interest in the "Research Assistant I (Student/Work Study)" position, where I will be tasked with designing and implementing a novel type system to integrate a practical subset of dependent types into a mainstream object-oriented context like Java or Kotlin and prevent runtime errors.

My recent experience as a SURE "AI in Education" Research Fellow at UM-Flint prepared me well for this opportunity. In that role, I analyzed academic literature, worked on developing a survey, and collaborated with faculty and peers from various schools to refine the direction of my research. Presenting findings in weekly discussions strengthened my ability to communicate technical concepts clearly and adapt my research goals based on feedback, which are skills that translate directly into collaborative software engineering projects.

In addition to my research experience, I bring strong technical skills in Java, Python, C++, and Lua, and I am eager to build familiarity with Kotlin and compiler design concepts. My freelance software development work has further trained me to deliver clean, modular code and to collaborate effectively in iterative, sprint-style settings.

This research assistantship excites me because it combines my passion for software engineering with open-source collaboration and faculty mentorship. I am eager to apply my skills to the challenge of advancing type system design while deepening my knowledge of compiler construction.

Thank you for taking the time to consider my application. I have attached my resume for your review and would welcome the opportunity to discuss how my background and interests align with the goals of this research opportunity. I look forward to hearing from you.

Sincerely,
Andrew Haralambous